



HK IMO Training 2019-2020
Problem Set 5
For 16 November 2019



Problem 17. Mandy and Samuel play the following game by making their moves alternately; Mandy moves first (yes, ladies first!). Initially they have a large piece of modelling clay. By a move, Mandy cuts one of the existing pieces into three (of arbitrary sizes), and Samuel merges two existing pieces into one. Mandy wins if at some moment there appears to be 100 pieces of equal weights. Can Samuel prevent Mandy from winning the game?

Problem 18. The rhombus $ABCD$ is given. Let E be one of the points of intersection of the circles Γ_B and Γ_C , where Γ_B is the circle with centre B and passes through C , and Γ_C is the circle with centre C and passes through B . The line ED intersects Γ_B at point F . Find the value of the angle $\angle AFB$.

Problem 19. Let p and q be prime numbers. Show that if $p + q^2$ is a perfect square then for any positive integer n the number $p^2 + q^n$ is not a perfect square.

Problem 20. There are n cards on the desk. Each of the n cards has one of the numbers from 1 to n and all numbers on the n cards are distinct. The cards are initially stacked so that the numbers on the cards increase from top to bottom. Let A be an action to move the top card of the stack to the bottom of the stack and B an action to remove the top card of the stack out of the desk. Repeat actions in the following order until the last one is left on the desk: ABBABBABBABB $\dots$.

Let $L(n)$ be the number on the last remaining card. Find all the positive integers k satisfying $L(3k) = k$.